Industrial Internship Report on

”Full Stack Food Delivery Application-DineLink”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 4 weeks’ time. My project was DineLink and it is a full-stack food delivery application developed as part of a Full Stack Development internship. The project focuses on building a platform that connects customers and restaurants, allowing users to explore restaurants while enabling restaurant owners to manage their services. The application is built using the MERN stack (MongoDB, Express.js, React.js, Node.js) and demonstrates a complete end-to-end system. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

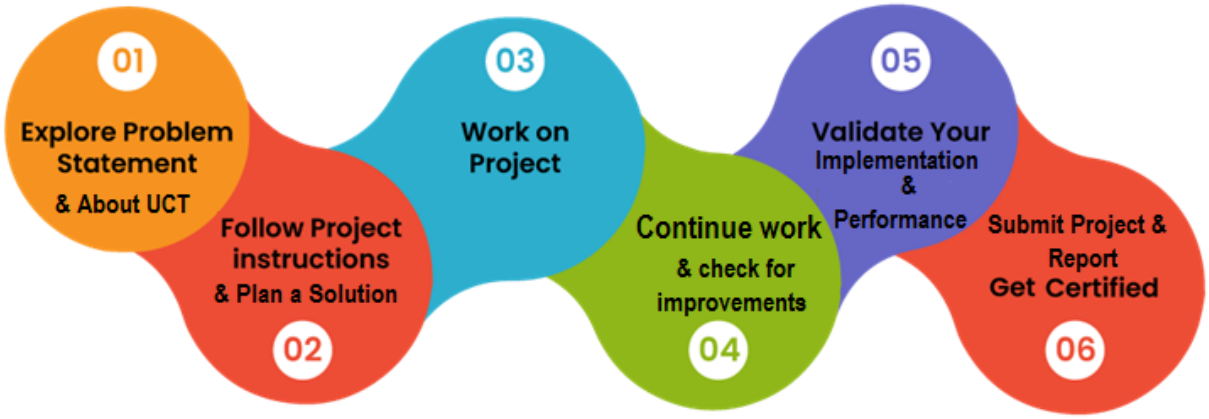
Summary of the whole 4 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

Thank to all , who have helped you directly or indirectly.

My message to your juniors and peers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

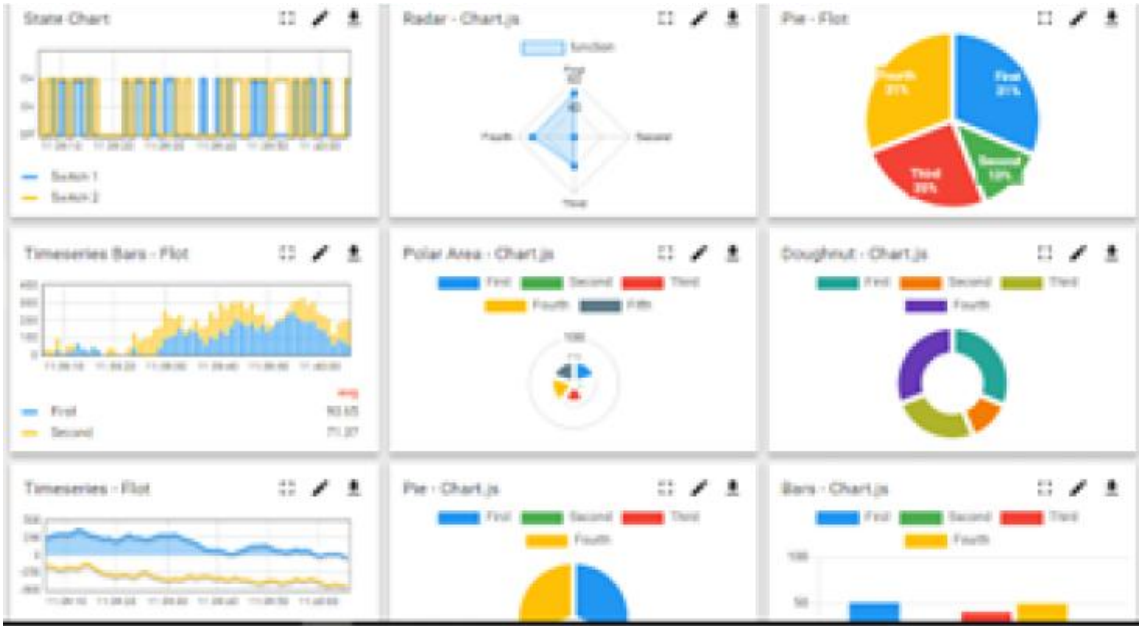
i. UCT IoT Platform (UCT Insight)

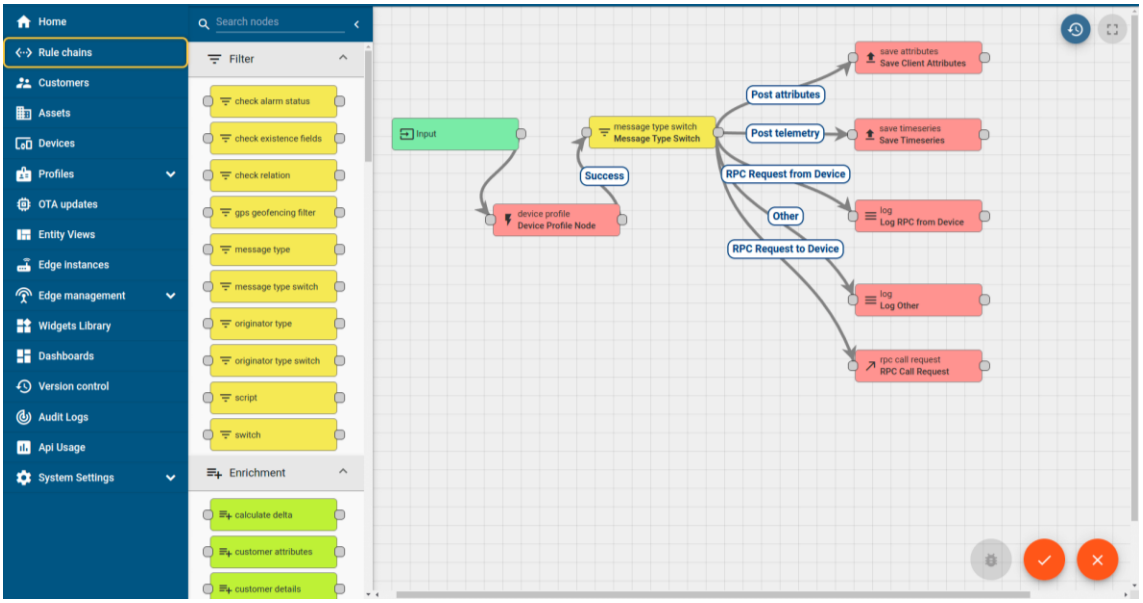
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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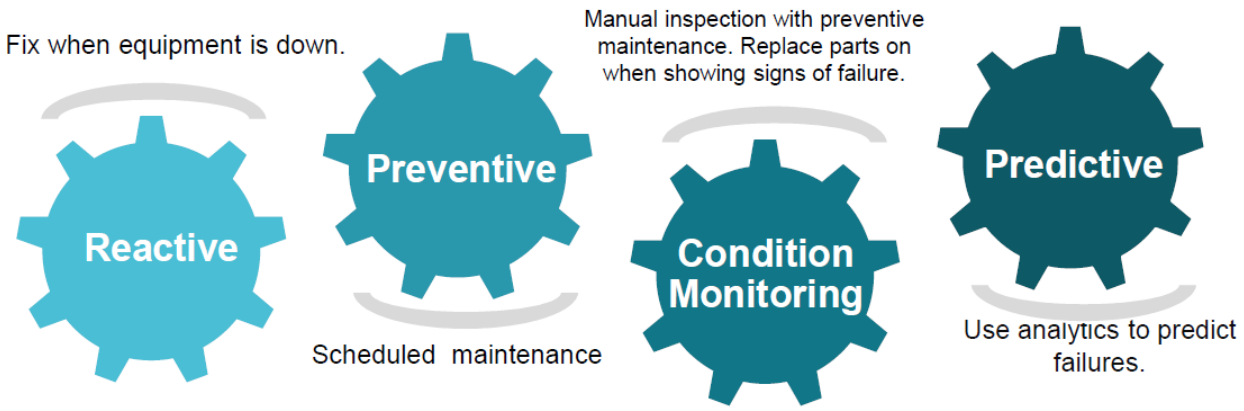
iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

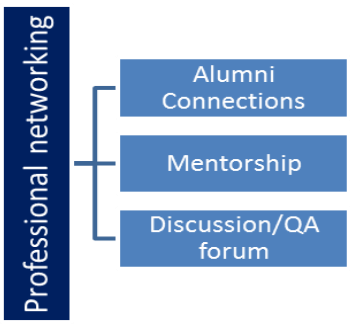
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.

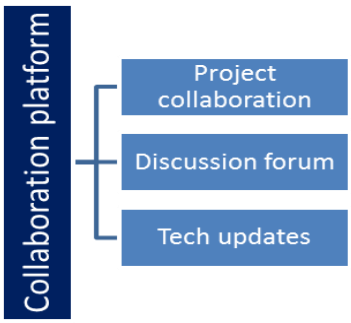


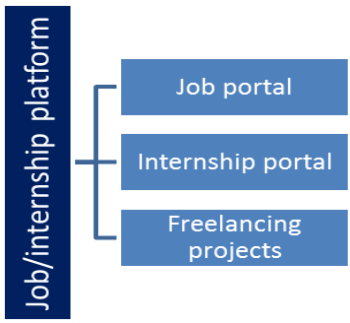
2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.







2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The main objectives of this project are:

- To develop a secure backend system
- To implement user authentication using JWT
- To enable role-based access control
- To build a frontend using React
- To integrate frontend and backend seamlessly.

2.5 Reference

[1] Node.js Documentation

[2] React.js Documentation

[3] MongoDB Documentation

[4] JWT Documentation

3 Problem Statement

In the assigned problem statement

Food ordering is still done in traditional ways which can be slow and sometimes confusing. There can be delays, wrong orders and long waiting times, especially when it is busy. Customers may find it inconvenient to order food and restaurants also find it hard to manage many orders properly. Also, these methods are not suitable for today's digital world where people expect fast and easy services. So, there is a need for a simple and efficient system to improve the food ordering process.

4 Existing and Proposed solution

Provide summary of existing solutions provided by others, what are their limitations?

Currently, many restaurants follow traditional methods like taking orders manually or through phone calls. In this system, there are chances of mistakes in orders, delays in processing, and confusion between customers and staff. During busy hours, it becomes difficult to manage multiple orders, which leads to long waiting times and poor customer experience. Also, customers do not get proper information about menu, availability, or order status.

What is your proposed solution?

The proposed system is the Dinelink application, which provides a simple and digital platform for food ordering. Through this app, users can easily browse menus, place orders, and view details without any confusion. It reduces human errors and saves time by making the process faster and more organized. Restaurants can also manage orders efficiently in one place. Overall, this system improves accuracy, convenience, and provides a better experience for both customers and restaurants.

What value addition are you planning?

In the future, the DineLink application can be improved by adding more useful features to enhance user experience. One of the planned additions is a real-time order tracking system, so users can know the status of their order. Another feature is online payment integration to make transactions faster and more convenient.

The app can also include a review and rating system, where customers can give feedback about restaurants and food. Personalized recommendations based on user preferences can also be added to make the app more user-friendly.

4.1 Code submission (Github link)

[https://github.com/Nihara-12/upskillCampus]

4.2 Report submission (Github link) :.

[https://github.com/Nihara-12/upskillCampus]

5 Proposed Design/ Model:

The DineLink application follows a client-server architecture where the frontend communicates with the backend through REST APIs, and the backend interacts with the database for storing and retrieving data.

The system is divided into three main layers:

- **Frontend (User Interface)**
- **Backend (Application Logic)**
- **Database (Data Storage)**

The workflow starts from user interaction on the frontend, passes through backend APIs for processing, and ends with data storage or retrieval from the database

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5.1 High Level Diagram (if applicable)

[React Frontend] → [Node.js / Express Backend] → [MongoDB Database]

The high-level design shows the interaction between the frontend, backend, and database. The frontend sends HTTP requests to the backend, which processes the request and interacts with MongoDB. The response is then sent back to the frontend.

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

User → React UI → Axios API Calls → Backend Routes →Controllers →Middleware (JWT Auth) →

Database (MongoDB).

5.3 Interfaces (if applicable)

The system uses REST APIs as the primary interface between the frontend and backend. The communication follows HTTP protocols.

Key Interfaces:

- POST /api/auth/register → User registration
- POST /api/auth/login → User authentication
- GET /api/restaurants → Fetch all restaurants
- POST /api/restaurants → Add restaurant (protected route)

Data Flow:

- Frontend sends JSON requests
- Backend processes requests and validates data
- MongoDB stores/retrieves data
- Response is sent back in JSON format

6 Performance Test

This is very important part and defines why this work is meant of Real industries, instead of being just academic project.

Here we need to first find the constraints.

How those constraints were taken care in your design?

What were test results around those constraints?

Constraints can be e.g. memory, MIPS (speed, operations per second), accuracy, durability, power consumption etc.

In case you could not test them, but still you should mention how identified constraints can impact your design, and what are recommendations to handle them.

6.1 Test Plan/ Test Cases

Constraints Identified

- API response time
- Database query performance
- Network latency
- Concurrent user handling

How Constraints Were Addressed

- Efficient API design using Express.js
- Use of MongoDB indexing for faster queries
- Lightweight JSON responses
- Middleware to optimize request handling

6.2 Test Procedure

- APIs were tested using command-line tools and browser-based testing
- Each endpoint was verified with valid and invalid inputs
- Authentication was tested using JWT tokens
- Database entries were checked in MongoDB Atlas

6.3 Performance Outcome

- The system successfully handled user authentication and data retrieval without delays
- API response time remained within acceptable limits for small-scale usage
- Data was stored and retrieved accurately from MongoDB
- The application demonstrated stable performance under normal usage conditions

Due to limited deployment environment, large-scale performance testing could not be conducted..

7 My learnings

- Gained hands on experience in full stack development using MERN stack
- Learned how authentication and authorization work in real world applications
- Improved debugging and problem solving skills
- Understood how frontend and backend systems communicate
- Built confidence in developing and deploying complete applications

Overall this Internship provided me with a great first time hands on real world application experience and I would really like to thank UCT Academy for providing me with this wonderful free Full Stack Application Internship program.

8 Future work scope

- Add food menu and order management system
- Implement payment gateway integration
- Add real-time delivery tracking
- Improve UI/UX design
- Deploy the application for public access

The DineLink project successfully demonstrates a working full stack application with authentication, role-based access, and restaurant management. The integration of frontend and backend provides a complete system.